Editor-in-Chief, IEEE Transactions on Intelligent Vehicles

Submission: Regular paper, "Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction"

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Dear Editor,

We submit the above manuscript for consideration as a Regular paper in IEEE Transactions on Intelligent Vehicles.

Conformal prediction regions are used to give trajectory forecasts a stated coverage, for example 90%, for safe planning around other road users. That guarantee assumes that calibration and deployment data are exchangeable, which a forecaster moved to a different driving dataset violates. The paper measures, on Argoverse 2, nuScenes and Waymo Open Motion and on two forecasting architectures, how much coverage is lost when no target labels are available (6.4 to 25.0 points at a 90% target), shows that covariate reweighting and the other label-free repairs we tested do not remove the loss, and gives a scene-level audit and repair protocol with finite-sample guarantees and measured label budgets. All hypotheses and success criteria were registered before each experiment, and negative results are reported.

The manuscript is original, has not been published, and is not under consideration elsewhere. The authors declare no conflict of interest, and the work received no external funding. The Acknowledgment discloses the use of generative AI (Anthropic's Claude) in preparing the article. Code, the registered hypotheses and the result files are available at https://doi.org/10.5281/zenodo.23054412 and https://github.com/hwaqas160/conformal-coverage-transfer.

Thank you for considering our work.

Sincerely, Hassan Waqas Saleem (corresponding author, hsaleem.msee20seecs@seecs.edu.pk) and Rahat Zeeshan Saleem